Week 4: Learning with Physical Constraints

Mismatch between predicted total and feed concentration

An MLP predicts [0.70, 0.45, 0.25] mol/L for A, B, and C with a 1.20 mol/L feed. The total is 1.40. Can an engineer pass this outlet to the next process unit?



Where did 0.20 mol/L come from?

Propagation of balance errors between process units

The next separator calculates its load from incoming component amounts. A total-concentration error in the reactor surrogate enters that calculation directly, so prediction error and balance violation need separate checks.

This week: residual penalties in the loss, output projection onto specified linear equalities, and component errors before and after correction.

Comparisons for learning with physical constraints

1. MLP, soft sum penalty, and hard sum projection on the same data
2. Nearest-point projection from KKT conditions; rank assumptions
3. Hand gradients versus automatic differentiation through a fixed layer
4. Separate reporting: component errors, equality violations, negative concentrations, and unenforced balances

Lab reference: the Week 2 steady-state CSTR.

Reference process and units

Assume perfect mixing, constant volume, and equal inlet/outlet flow. The feed contains only A; consecutive reactions A → B → C have 1:1 stoichiometry. Temperature T is prescribed and the model has no energy balance.

$$C_A = \frac{C_{\mathrm{Af}}}{1 + k_1\tau}$$

$$C_B = \frac{k_1\tau C_A}{1 + k_2\tau}, C_C = k_2\tau C_B$$

At 330 K, $k_1 = 0.2\ \mathrm{min}^{-1}$ and $k_2 = 0.1\ \mathrm{min}^{-1}$. With $\tau = 5$ min and C_Af = 1.2 mol/L, [C_A, C_B, C_C] = [0.6, 0.4, 0.2] mol/L.

Steady-state balances and the concentration-sum equality

Let r_A, r_B, and r_C denote the right sides of the steady-state species balances. Adding them cancels the reaction terms.

$$r_A + r_B + r_C = \frac{C_{\mathrm{Af}} - \left(C_A + C_B + C_C\right)}{\tau}$$

At steady state each residual is zero, giving S = C_A + C_B + C_C = C_Af. This concentration-sum equality depends on the stated stoichiometry and flow assumptions. Other reactions require the appropriate elemental or mass balance and units.

Componentwise prediction error and concentration-sum error

Prediction [mol/L]	Component error [mol/L]	Sum residual [mol/L]
[0.70, 0.45, 0.25]	[+0.10, +0.05, +0.05]	+0.20
[0.70, 0.30, 0.20]	[+0.10, −0.10, 0.00]	0.00

The reference is [0.60, 0.40, 0.20] mol/L. The second prediction has the right sum but incorrect A and B. An equality-only evaluation would miss those errors.

Definition of physical residuals

The sum residual is r_sum = C_A + C_B + C_C − C_Af, in mol/L. Known kinetics also allow species-level steady-state residuals.

$$r_A = \frac{C_{\mathrm{Af}} - C_A}{\tau} - k_1 C_A$$

$$r_B = k_1 C_A - \frac{C_B}{\tau} - k_2 C_B$$

$$r_C = k_2 C_B - \frac{C_C}{\tau}$$

Units for these three residuals: mol/(L·min). Training: sum residual only. Separate evaluation: species residuals.

Data-error and constraint-violation terms in a soft penalty

$$L_{\text{soft}} = \frac{1}{3N}\sum_i \left\| \frac{\hat{y}_i - y_i}{c_*} \right\|^2 + \lambda \frac{1}{N}\sum_i \left(\frac{r_{\text{sum},i}}{c_*} \right)^2$$

Both terms are dimensionless with c_* = 1 mol/L. The lab uses λ = 10; its meaning includes the averaging conventions for one sum residual and three component errors.

Changing λ changes the objective and gradient magnitudes. A finite penalty and a completed training run do not guarantee zero residual at every input.

Solve a finite-penalty problem by hand

Minimize the scalar objective ½(z − a)² + ½λ(z − b)², where data favor a, the equality requires z = b, and λ ≥ 0. Setting the derivative to zero gives:

$$z_* = \frac{a + \lambda b}{1 + \lambda}$$

$$z_* - b = \frac{a - b}{1 + \lambda}$$

If a ≠ b, a residual remains for finite λ. Very large penalties also require attention to loss magnitudes and numerical conditioning.

Differentiation targets in automatic differentiation

Our MLP maps [T, τ, C_Af] to steady concentrations. Training differentiates the loss with respect to parameters θ. The sum residual and steady-state species balances need no time derivative.

A continuous-time PINN using $y_\theta(t)$ may have a residual $\partial y_\theta/\partial t - F(y_\theta, u)$, which also differentiates with respect to a coordinate. Distinguish this from the parameter gradient used for training.

If $\tilde{t} = (t - \mu_t)/s_t$, then $\partial y/\partial t = (1/s_t)\partial y/\partial \tilde{t}$.

Hard linear equalities and feasible output sets

$$Ax + By = b,$$
$$By = d(x), d(x) = b - Ax$$

$x \in \mathbb{R}^p$, $y \in \mathbb{R}^n$, $A \in \mathbb{R}^{mxp}$, $B \in \mathbb{R}^{mxn}$, and $b \in \mathbb{R}^m$. Coefficients A, B, b are fixed; the raw output is $\hat{y} = f_\theta(x)$.

For the CSTR, $x = [T, \tau, C_{Af}]$, $A = [0, 0, -1]$, $B = [1, 1, 1]$, and $b = 0$. Thus $d(x) = C_{Af}$.

Nearest-point projection onto a linear equality set

$$y_p = \text{argmin}_z \frac{1}{2} \left\| z - \hat{y} \right\|^2 \text{subject to} Bz = d(x)$$

Fixed quantities: x and raw prediction ŷ. Adjusted variable: output z. Metric: Euclidean distance in physical concentration coordinates.

The objective is strictly convex, so a nonempty feasible set has a unique projected output. KKT conditions give a closed form for this equality-only QP.

Lagrangian and KKT conditions

$$\mathcal{L}(z,\lambda) = \frac{1}{2}\left\|z-\hat{y}\right\|^2 + \lambda^T(Bz-d)$$

Enforce stationarity and primal feasibility together.

$$z-\hat{y}+B^T\lambda = 0, Bz = d$$

There are no inequalities, hence no complementary slackness or multiplier sign restriction. For a strictly convex objective with affine equalities, an output satisfying these conditions is the global minimizer.

Eliminate the multiplier to obtain the projection

Stationarity gives $z = \hat{y} - B^T\lambda$. Substitute into $Bz = d$.

$$G = BB^T, G\lambda = B\hat{y} - d$$
$$y_p = \hat{y} - B^T G^{-1}\left(B\hat{y} - d\right)$$

If rank(B) = m, then $G = BB^T$ is positive definite. The formula uses an inverse for notation; implementation solves $G\lambda = B\hat{y} - d$. A fixed B allows reuse of a factorization or fixed projection coefficients.

The fixed affine output layer in KKT-hPINN

$$P = I - B^T G^{-1} B$$
$$y_p = P f_\theta(x) + B^T G^{-1}(b - Ax)$$

Physical equalities determine this layer's coefficients; they are not trained. MLP weights θ are trained using the projected output in the loss.

$$B y_p = B\hat{y} - G G^{-1}\left(B\hat{y} - d\right) = d$$

The specified equalities hold during training and inference, subject in implementation to floating-point error and linear-system conditioning. Source: Chen et al. (2024).

Constraint-matrix rank and equality-system consistency

Case	Consequence
rank(B) = m ≤ n	Every d is attainable; BB^T is invertible.
Redundant rows, consistent d	Feasible outputs exist but BB^T is singular. Use independent rows or an SVD.
Redundant rows, inconsistent d	No feasible output exists. A pseudoinverse cannot satisfy all equations.

For nearly dependent rows, inspect rank tolerance and conditioning. Scaling entire equality rows consistently changes numerical scales without changing the feasible set.

Project the concentration sum by hand

$$B = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, d = 1.2$$

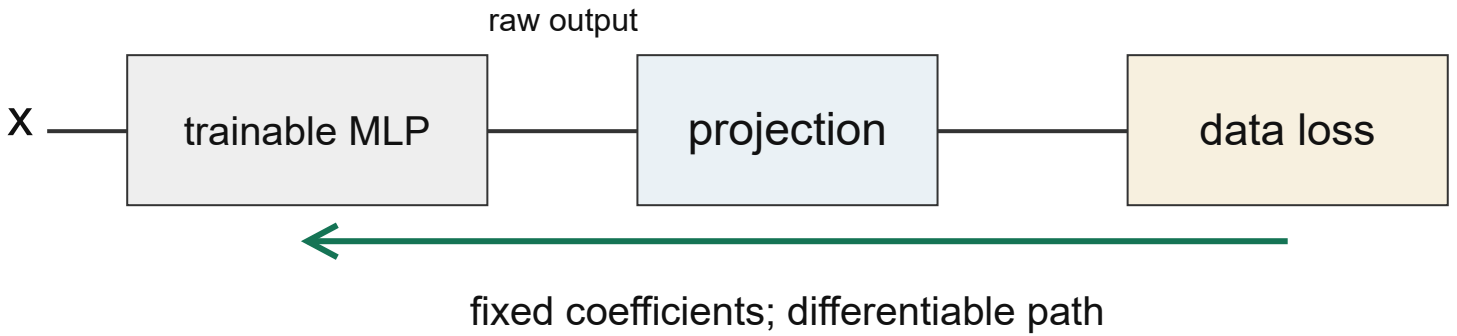
$$y_p = \begin{bmatrix} .70 \\ .45 \\ .25 \end{bmatrix} - \frac{.20}{3}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} .633333 \\ .383333 \\ .183333 \end{bmatrix}$$

Subtract the same 0.066667 mol/L from each component. The total is 1.20, but the output still differs from the reference [0.60, 0.40, 0.20].

Displayed values are rounded. Keep full precision during calculation and check the final sum residual.

Post-projection and training through projection

Method	Output used in the training loss
MLP + post-projection	Train on raw outputs; project at test time.
KKT-hPINN approach	Use projected outputs throughout training.



Both can satisfy the same equality, but their training gradients differ and generally produce different parameters.

Backpropagation through a fixed projection layer

$$\frac{\partial y_p}{\partial \hat{y}} = P$$

$$\nabla_{\hat{y}} L = P^T \nabla_{y_p} L$$

P is a symmetric orthogonal projector, so $P^T = P$. The projection matrix P is fixed and passes the output gradient to the raw prediction by multiplying it by P^T.

Detaching the output or wrapping projection in no_grad cuts the required path. The parameter gradient continues through the MLP Jacobian by the chain rule.

Check the backward pass with three components

$$P = I - \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ \frac{-1}{3} \\ \frac{-1}{3} \end{bmatrix}$$

An upstream gradient [1, 0, 0] at the projected output becomes [2/3, −1/3, −1/3] at the raw output. Its sum is zero, so it lies in the equality's tangent space.

Additional checks: analytic Jacobian and double-precision gradcheck for two independent equalities.

Constraint-matrix transformation under output standardization

$$y = \mu + Dz, BDz = d - B\mu$$
$$D = \mathrm{diag}\left(s_1, \ldots, s_n \right)$$

Compute μ and positive scales s_j from training data only. A projection in standardized coordinates z must use BD and right side d − Bμ.

Projection in standardized coordinates minimizes a distance in physical coordinates that weights each squared component correction by $1/s_j^2$. Specify the correction metric and scales.

Physical-unit restoration and projection order

The MLP returns standardized outputs z. Restore mol/L with μ + Dz, then project in physical coordinates. Divide data errors by the common scale c_* = 1 mol/L.

$$\nabla_z L = D^T P^T \nabla_{y_p} L$$

D is diagonal, hence $D^T = D$. A loss divided by component-specific standard deviations adds weights that must also enter gradient and error interpretations.

A minimal PyTorch implementation

raw units: mol/L. feed shape: (batch, 1).

```
raw = y_mean + y_scale * net(x_scaled)
residual = raw.sum(dim=-1, keepdim=True) - feed
y_projected = raw - residual / 3.0
loss = ((y_projected - y_true) / c_scale).square().mean()
optimizer.zero_grad()
loss.backward()
optimizer.step()
```

Projection adds no trainable parameters. The companion and lab script include the general B implementation and gradient checks.

Concentration-sum constraints and nonnegativity

Raw [0.02, 0.90, 0.50] mol/L sums to 1.42. Projection to 1.20 gives [−0.053333, 0.826667, 0.426667] mol/L.

The sum is correct and A is negative. Clamping just the negative value to zero changes the sum to about 1.253333.

To require both nonnegativity and the sum, use a method including those inequalities, such as simplex projection. Its gradient changes by region as active inequalities change.

Conditions for error reduction by projection

If reference y^* lies in the same affine set, orthogonality gives:

$$\left\|\hat{y} - y^*\right\|^2 = \left\|\hat{y} - y_p\right\|^2 + \left\|y_p - y^*\right\|^2$$

Thus physical-space projection of the same raw prediction cannot increase total squared Euclidean error. An individual component error can increase.

If noisy labels lie outside the set, or projection uses another weighted metric, this physical Euclidean comparison does not directly apply. The identity also does not rank two separately trained models.

Fix the comparison protocol

Use $T = 305\text{-}355$ K, $\tau = 1\text{-}8$ min, $C_{Af} = 0.8\text{-}1.5$ mol/L. Rates are $k_1 = 0.2 \exp[6000(1/330 - 1/T)]$ and $k_2 = 0.1 \exp[5000(1/330 - 1/T)]$ min^{-1}.

Train/validation/test have 900/300/300 rows, seed 404, and noise-free analytic labels. All models are $3 \rightarrow 32 \rightarrow 32 \rightarrow 3$ ReLU MLPs with 1283 parameters.

They share initial weights, batch order, and training statistics. Use Adam 0.001, batches of 128, and the minimum validation physical MSE among 600 epochs; then evaluate test.

Run results: accuracy and equality violations

Model	Pooled RMSE	Max sum violation	Negative entries
MLP	0.003657	1.17e-02	1
Soft ($\lambda = 10$)	0.008161	6.30e-03	5
MLP + post	0.003426	2.22e-16	1
Hard	0.003036	3.33e-16	1

Concentration and error units: mol/L. Negative-entry count: values below -10^{-8} among 900 test component values.

In this run, the soft penalty reduced sum violations but increased RMSE. Hard and post satisfied the sum while retaining negative concentrations. One seed does not establish a general accuracy ranking.

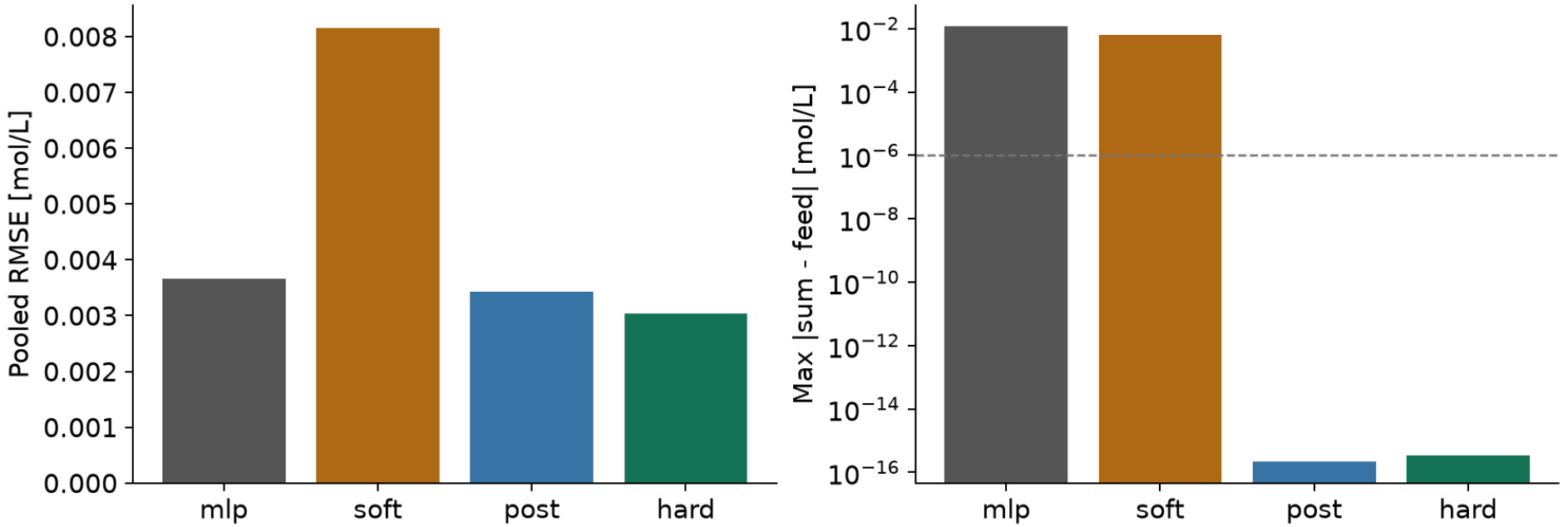
Component RMSE and minimum concentration

Model	A RMSE	B RMSE	C RMSE	Minimum
MLP	0.003866	0.003383	0.003704	-0.003451
Soft ($\lambda = 10$)	0.008753	0.008742	0.006840	-0.010393
MLP + post	0.003724	0.003015	0.003500	-0.004176
Hard	0.003728	0.002540	0.002703	-0.002822

All quantities are in mol/L. Pooled RMSE can hide differences between components.

The minimum covers every component and test row. Record the evaluation tolerance and the actual minimum together.

Evaluation of species balances omitted from training



The hard model satisfies the concentration-sum equality, while its component reaction/outlet relations retain a residual RMS of 0.002225 mol/(L·min).

Run the lab and retain the evidence

Download the lab ZIP, notebook, or Python script. Extract the ZIP and run in that folder.

```
python -m pip install -r requirements.txt
python week04_lab.py --check-only
python week04_lab.py --output-dir week04-results
```

Outputs retain the split dataset, test predictions, weights, training statistics, validation history, and evaluation JSON. The supplied run used CPU, float64, and PyTorch 2.14.1+cpu.

Total-concentration equality in the dynamic model

With constant feed and τ within an interval, the exact total-concentration transition is:

$$S_{k+1} = C_{\mathrm{Af}} + \left(S_k - C_{\mathrm{Af}}\right) e^{-\frac{\Delta t}{\tau}}$$

If S_k differs from C_Af, do not immediately force the next total to C_Af. B = [1, 1, 1] still acts on the next state, but the right side is the transition above.

When τ is an input, the equality right side d(x) depends nonlinearly on τ. Compute that right side, then apply the projection with fixed B.

Exercises and submission

1. Project [0.70, 0.45, 0.25] onto total 1.20; backpropagate an upstream gradient [1, 0, 0].
2. Explain feasibility for B rows [1, 1, 1], [2, 2, 2] and d = [1.2, 2.5].
3. Compare $\lambda = 0.1, 1, 10$ using validation; evaluate only the selected setting on test.
4. Submission for each model: A/B/C RMSE, sum MAE/max, minimum concentration, and species-balance residuals

Next week: frozen ReLU predictor and optimization-problem definition.

Reading companion and papers

Week 4 companion · English reading PDF

Chen, H., Constante Flores, G. E., & Li, C. (2024). *Physics-informed neural networks with hard linear equality constraints*. Computers & Chemical Engineering, 189, 108764. DOI · Public manuscript · Author code/data.

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). *Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations*. Journal of Computational Physics, 378, 686-707. DOI.

Mathematics and implementation sources

Boyd, S., & Vandenberghe, L. (2004). *Convex Optimization*. Section 5.5.3 on KKT conditions and Section 10.1 on equality-constrained minimization. Official book.

Official PyTorch documentation: linear solves, numerical gradient checks. Raissi et al., 2017 Part I manuscript.